

Tutorial 2 #5 table of equivalences

for linear map $f: \mathbb{R}^c \rightarrow \mathbb{R}^r$
 $x \mapsto Ax$

Rmk # pivots $\text{REF}(A) \leq \min(r, c)$

$$\left. \begin{array}{l} \textcircled{\text{I}} \Rightarrow r \leq c \\ \textcircled{\text{II}} \Rightarrow r \geq c \end{array} \right\} \textcircled{*}$$

f

system of eqⁿ

Cols of A

Rows of A

$\text{REF}(A)$

$\textcircled{\text{I}}$ f surjective $\Leftrightarrow$ For each $b \in \mathbb{R}^r$
 $Ax=b$ has a solⁿ $\Leftrightarrow$ Cols of A span $\mathbb{R}^r$ $\Leftrightarrow$ Rows of A are lin. indep $\Leftrightarrow$ $\text{REF}(A)$ has a pivot in each row.

$\textcircled{\text{II}}$ f injective $\Leftrightarrow$ $Ax=0$ has exactly one solⁿ, namely $x=0$
 ("trivial" solⁿ) $\Leftrightarrow$ Cols of A are linearly indep $\Leftrightarrow$ Rows of A span $\mathbb{R}^c$ $\Leftrightarrow$ $\text{REF}(A)$ has a pivot in each col

Must be able to justify entire table with ease.

A vector space has well behaved $+$, $-$, $\cdot$.

Words: linear combination, span, linear (in)dependence, basis, dimension

Defⁿ V is "finite dimensional" means $V = \text{span}(\text{some finite set } S)$. $\downarrow$

Observe A "fdvsp" has a basis.

by using strategy (B).

Q. Will strategy (A) also succeed always for fdvsp.

Q. Why is $\dim$ of any fdvsp well-defined?

Two strategies to get a basis of any V

(A) keep adding elts, preserving lin independence.

(B) keep deleting elts from a spanning set, preserving span

Termination? problem for both!

Observe $S \subset V \cdot \text{sp } V$

S is dependent $\iff$ some vector v in S is in the span of $S \setminus \{v\}$

S is independent and $w \notin \text{span}(S) \Rightarrow S \cup \{w\}$ is independent.

✓ A maximal independent set in V is a basis of V .
✓ A minimal spanning set in V is a basis of V .

"maximal/minimal" means
with respect to set
containment

Rmk. Let $\{v_1, \dots, v_k\} = S \subset \mathbb{R}^n$. S is lin indep $\Rightarrow n \geq k$
Take $A = \text{matrix with } v_i \text{ as cols}$ $S \text{ spans } \mathbb{R}^n \Rightarrow k \geq n$

$\therefore \text{Dim of } \mathbb{R}^n \text{ is } n$

$\Rightarrow n = k$

How to find a basis for (For given $r \times c$ matrix A)

① $\text{Col}(A) = \text{range}(x \mapsto Ax) \subset \mathbb{R}^r$ → Easy check that they are indep

② $\text{Nul}(A) = \{x \mid Ax=0\} \subset \mathbb{R}^c \rightsquigarrow$ magic vectors obtained using $\text{RREF}(A)$

③ $\text{Row}(A) = \text{span of row vectors of } A \subset \mathbb{R}^c$

row opⁿ do not change the row space!

Ex: A basis for $\text{row}(A)$ is $\{\text{nonzero rows of } \text{REF}(A)\}$

Ex Row opⁿ do change the column space. So how do I solve ①?

Ex.

Suppose we have

linear $f: V \rightarrow W$

with $w \in \text{range}$, say $w=f(v)$

Construct bijection

$$\ker(f) = \tilde{f}^{-1}(0) \longleftrightarrow \tilde{f}^{-1}(w)$$

In fact, express $\tilde{f}^{-1}(w)$ in terms of $\ker(f)$